

Software Engineering Project: Offline Accessible Multimodal Local Content Retrieval System

Duration: 8 Weeks (2 Months)

Project Background

This project aligns with our founding mission: *to organize the world's information and make it universally accessible and useful*. It replicates the end-to-end software engineering workflow of core product teams, focusing on solving a real user pain point: billions of users including people with visual impairments are struggling to efficiently search and retrieve unstructured local content across personal devices, such as PDFs, documents, images, screenshots. Existing solutions are often cloud-dependent, paid, lack robust accessibility features, or fail to support semantic multimodal search.

Core Project Goals

1. Build a fully open-source, offline-first, cross-platform multimodal local content retrieval tool
2. Meet WCAG 2.1 AA accessibility standards, aligned with global accessibility commitments
3. Deliver production-grade code, full documentation, and a deployable end product following software engineering best practices

Core Tech Stack

- Cross-platform UI: Flutter
- ML Inference: TensorFlow Lite (end-side inference framework)
- Multimodal Models: BERT (text embedding) + MobileCLIP (image embedding)
- Document Parsing: PDFium (Chromium PDF engine) + Apache Tika
- Vector Storage: Chroma DB (local vector database)
- Testing: Google Test + Flutter Test
- Version Control: Git + GitHub
- Accessibility Validation: Google Accessibility Scanner + WAVE

Weekly Project Plan

Week 1: Project Onboarding, Requirement Definition & Environment Setup

Core Focus: Align with engineering standards, finalize scope, and build a fully functional development environment

Key Tasks	Weekly Deliverables
1. Review software engineering best practices, accessibility guidelines, and open-source compliance rules2. Finalize functional/non-functional requirements (offline-first, cross-platform support, WCAG 2.1 AA compliance)3. Set up full development environment: Flutter SDK, TensorFlow Lite, Git, Chroma DB, and all open-source dependencies4. Download and curate validation splits of required datasets5. Complete project risk assessment and mitigation plan	1. Signed-off Project Requirements Document (PRD)2. Environment setup validation report3. Curated local dataset repository4. Project risk management plan

Week 2: System Architecture Design & Core File Parsing Module Development

Core Focus: Design a modular, maintainable architecture aligned with Google’s clean code principles, build the foundation of the data ingestion pipeline

Key Tasks	Weekly Deliverables
1. Design full modular system architecture: File I/O Layer, Parsing Layer, Embedding Engine Layer, Vector Storage Layer, Retrieval Logic Layer, UI & Accessibility Layer2. Develop and unit-test the file parsing module (support for TXT, PDF, DOCX, JPG, PNG formats)3. Implement batch file ingestion and metadata extraction logic4. Draft full technical design document (TDD) with architecture diagrams	1. Finalized System Architecture Design Document2. Functional file parsing module with ≥80% unit test coverage3. Module API documentation

Week 3: Multimodal Embedding Engine Development

Core Focus: Integrate models to build a local, offline embedding engine for text and image content

Key Tasks	Weekly Deliverables
1. Optimize and integrate BERT-base model for text embedding, converted for TensorFlow Lite offline CPU inference2. Integrate MobileCLIP model for lightweight image embedding, optimized for end-side execution3. Build a unified embedding interface for standardized text/image input processing4. Validate embedding accuracy using the NQ and COCO validation datasets5. Implement batch embedding processing for large local file libraries	1. Functional multimodal embedding engine2. Model accuracy validation report3. Unit test suite for the embedding module (≥85% test coverage)

Week 4: Vector Database Integration & Core Retrieval Logic Implementation

Core Focus: Build the end-to-end semantic search pipeline, deliver a working minimum viable product (MVP)

Key Tasks	Weekly Deliverables
1. Integrate Chroma DB for local vector storage, indexing, and similarity search2. Develop hybrid semantic retrieval logic: keyword + embedding-based similarity scoring, result ranking, and filter functionality3. Build the end-to-end retrieval pipeline: file ingestion → parsing → embedding → storage → search → result return4. Conduct end-to-end functional testing of the core pipeline5. Benchmark retrieval accuracy using curated test datasets	1. Integrated vector storage and retrieval module2. Working MVP of the full core retrieval pipeline3. End-to-end functional test report4. Retrieval accuracy benchmark report

Week 5: Cross-Platform UI Development & Accessibility Feature Implementation

Core Focus: Build a user-facing interface, implement industry-leading accessibility features

Key Tasks	Weekly Deliverables
1. Develop cross-platform UI with Flutter, including core screens: file library management, search interface, result display, and settings2. Implement mandatory accessibility features: screen reader support (NVDA/VoiceOver), keyboard-only navigation, high-contrast mode, dynamic font scaling3. Validate accessibility compliance with Google's Accessibility Scanner and WAVE tool4. Conduct usability testing for core UI and accessibility workflows	1. Functional cross-platform Flutter UI2. Accessibility compliance validation report (WCAG 2.1 AA aligned)3. UI usability test report4. Draft accessibility user guide

Week 6: System Integration, Full Testing & Performance Optimization

Core Focus: Stabilize the full application, ensure production-grade reliability, and optimize performance for consumer hardware

Key Tasks	Weekly Deliverables
1. Full end-to-end integration of all modules (parsing, embedding, retrieval, UI)2. Execute comprehensive test plan: unit tests, integration tests, end-to-end tests, and stress tests for large file libraries3. Optimize performance: reduce embedding inference latency, speed up vector search, and minimize memory footprint4. Fix all critical and high-priority bugs identified during testing5. Conduct a local data security review to ensure offline-only, no data leakage	1. Fully integrated stable application build2. Full test coverage report (≥90% unit test coverage for core modules)3. Performance optimization benchmark report4. Bug fix and security review report

Week 7: Documentation, Open Source Compliance & Final Polish

Core Focus: Complete production-grade documentation and finalize user experience

Key Tasks	Weekly Deliverables
1. Write full technical documentation: architecture deep dive, API references, and maintenance guide 2. Create a comprehensive end-user manual, including installation guide, feature walkthrough, and dedicated accessibility operation guide3. Complete open-source compliance review: verify all dependency licenses, prepare Apache 2.0 license for the project4. Implement final UI/UX polish and minor bug fixes5. Prepare project demo materials	1. Complete technical documentation suite2. End-user installation and operation manual3. Open source compliance report (aligned with OSS policies)4. Project demo script

Week 8: Final Delivery, Demo & Portfolio Packaging

Core Focus: Deliver the final production-grade product

Key Tasks	Weekly Deliverables
1. Build production release executables for Windows, macOS, and Linux2. Host the full open-source project on GitHub, with complete README, license, and contribution guidelines 3. Record a 5-minute project demo video showcasing core features, accessibility functions, and technical implementation4. Package all internship deliverables into a professional portfolio	1. Final cross-platform production release build2. Public GitHub repository with full production codebase3. Project demo video4. Full internship deliverable portfolio

Full Deliverables List

1. Production-ready, cross-platform, offline multimodal local content retrieval application
2. Full production-grade codebase hosted on GitHub, following Google’s clean code and engineering best practices
3. Complete technical documentation suite: system architecture, API references, and maintenance guide
4. End-user manual with dedicated accessibility usage guide
5. Comprehensive test suite with ≥90% core module unit test coverage, plus full integration and E2E test reports
6. Open source compliance report aligned with open-source policies
7. 5-minute project demo video

Datasets

Dataset Name	Primary Use Case	Official Download Link
Natural Questions (NQ)	Text embedding accuracy validation, long-document retrieval benchmarking	https://ai.google.com/research/NaturalQuestions
COCO (Common Objects in Context)	Image embedding accuracy validation, multimodal retrieval testing	https://cocodataset.org/#download
RVL-CDIP	Scanned document/image retrieval testing, OCR + embedding validation	https://www.cs.cmu.edu/~aharley/rvl-cdip/
Wikipedia Text Corpus	Long-form document retrieval benchmarking, batch processing performance testing	https://dumps.wikimedia.org/

Reference Materials & Learning Resources

Engineering & Compliance Guidelines

- Software Engineering Best Practices: <https://google.github.io/eng-practices/>
- Material Design 3 Guidelines: <https://m3.material.io/>

Core Technology Documentation

- Flutter SDK Official Documentation: <https://docs.flutter.dev/>
- TensorFlow Lite Official Guide: <https://www.tensorflow.org/lite/guide>
- BERT Official Repository: <https://github.com/google-research/bert>
- MobileCLIP Official Repository: <https://github.com/apple/ml-mobileclip>
- PDFium (Chromium PDF Engine) Documentation: <https://pdfium.googlesource.com/pdfium/>

Supporting Open-Source Tool Documentation

- Chroma DB Official Guide: <https://docs.trychroma.com/>
- Apache Tika Documentation: <https://tika.apache.org/>
- Google Test Framework Guide: <https://google.github.io/googletest/>
- WCAG 2.1 AA Accessibility Standard: <https://www.w3.org/TR/WCAG21/>

Learning Courses

- Machine Learning Crash Course: <https://developers.google.com/machine-learning/crash-course>
- Flutter & Dart Development Course: <https://developers.google.com/learn/topics/flutter>
- Web Accessibility Course: <https://web.dev/learn/accessibility/>